

Matrices & Matrix Algebra

ANSWER KEY



Matrix addition and scalar multiplication

- 1 Let A have rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and B have rows $\langle 0, -1 \rangle, \langle 2, 1 \rangle$. Find $A + B$ and $2A - B$.
 $A + B$ has rows $\langle 1, 1 \rangle, \langle 5, 5 \rangle$. $2A$ has rows $\langle 2, 4 \rangle, \langle 6, 8 \rangle$, so $2A - B$ has rows $\langle 2, 5 \rangle, \langle 4, 7 \rangle$.
- 2 Let C have rows $\langle 2, 0, -1 \rangle, \langle 1, 3, 4 \rangle$. Find $-3C$.
 $-3C$ has rows $\langle -6, 0, 3 \rangle, \langle -3, -9, -12 \rangle$.

Matrix multiplication

- 3 Let A have rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and B have rows $\langle 2, 0 \rangle, \langle 1, 3 \rangle$. Find AB .
Row 1 of AB : $\langle 1(2) + 2(1), 1(0) + 2(3) \rangle = \langle 4, 6 \rangle$. Row 2: $\langle 3(2) + 4(1), 3(0) + 4(3) \rangle = \langle 10, 12 \rangle$. So AB has rows $\langle 4, 6 \rangle, \langle 10, 12 \rangle$.
- 4 For a 2×3 matrix A and a 3×2 matrix B , what are the dimensions of AB ? Is BA defined, and if so, what dimension?
 AB is 2×2 (outer dimensions of A 's rows and B 's columns). BA is also defined here (inner dimensions $2 = 2$ match), giving a 3×3 matrix — note AB and BA aren't even the same SIZE in general, let alone equal.
- 5 Let A have rows $\langle 1, 0, 2 \rangle, \langle -1, 3, 1 \rangle$ (a 2×3 matrix) and B have rows $\langle 2, 1 \rangle, \langle 0, 1 \rangle, \langle 3, -1 \rangle$ (a 3×2 matrix). Find AB .
Row 1 of AB : $\langle 1(2) + 0(0) + 2(3), 1(1) + 0(1) + 2(-1) \rangle = \langle 8, -1 \rangle$. Row 2:
 $\langle -1(2) + 3(0) + 1(3), -1(1) + 3(1) + 1(-1) \rangle = \langle 1, 1 \rangle$. So AB has rows $\langle 8, -1 \rangle, \langle 1, 1 \rangle$.

Matrix multiplication is not commutative

- 6 For A, B above (rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and $\langle 2, 0 \rangle, \langle 1, 3 \rangle$), compute BA and compare it to AB .
Row 1 of BA : $\langle 2(1) + 0(3), 2(2) + 0(4) \rangle = \langle 2, 4 \rangle$. Row 2: $\langle 1(1) + 3(3), 1(2) + 3(4) \rangle = \langle 10, 14 \rangle$. So BA has rows $\langle 2, 4 \rangle, \langle 10, 14 \rangle$ — different from AB 's rows $\langle 4, 6 \rangle, \langle 10, 12 \rangle$, confirming matrix multiplication is not commutative in general.
- 7 Give one situation where $AB = BA$ IS guaranteed, even though matrix multiplication is not commutative in general.
If $B = I$ (the identity matrix) or, more generally, if $A = B$ (a matrix always commutes with itself), then $AB = BA$. Also, diagonal matrices of the same size always commute with each other.

The identity matrix and matrix inverses

- 8 Describe the 2×2 and 3×3 identity matrices by their rows.
 I_2 : rows $\langle 1, 0 \rangle, \langle 0, 1 \rangle$. I_3 : rows $\langle 1, 0, 0 \rangle, \langle 0, 1, 0 \rangle, \langle 0, 0, 1 \rangle$ — a 1 down the main diagonal, 0 everywhere else.
- 9 Verify that $AI_2 = A$ for A with rows $\langle 3, 5 \rangle, \langle 2, 1 \rangle$.
 Row 1: $\langle 3(1) + 5(0), 3(0) + 5(1) \rangle = \langle 3, 5 \rangle$. Row 2: $\langle 2(1) + 1(0), 2(0) + 1(1) \rangle = \langle 2, 1 \rangle$. So AI_2 has rows $\langle 3, 5 \rangle, \langle 2, 1 \rangle = A$, confirmed.

Invertibility of 2x2 matrices

- 10 For A with rows $\langle a, b \rangle, \langle c, d \rangle$, state the formula for A^{-1} .
 A^{-1} is $\frac{1}{ad - bc}$ times the matrix with rows $\langle d, -b \rangle$ and $\langle -c, a \rangle$, provided $ad - bc \neq 0$.
- 11 Find the inverse of A with rows $\langle 3, 1 \rangle, \langle 2, 1 \rangle$.
 $\det A = 3(1) - 1(2) = 1$. A^{-1} has rows $\langle 1, -1 \rangle, \langle -2, 3 \rangle$. (Check: multiplying A by this gives I_2 .)
- 12 Find the inverse of A with rows $\langle 2, 5 \rangle, \langle 1, 3 \rangle$.
 $\det A = 2(3) - 5(1) = 1$. A^{-1} has rows $\langle 3, -5 \rangle, \langle -1, 2 \rangle$.
- 13 Use the inverse found above to solve the system $2x + 5y = 7, x + 3y = 4$ (writing it as $A\mathbf{v} = \mathbf{b}$ and computing $\mathbf{v} = A^{-1}\mathbf{b}$).
 A^{-1} has rows $\langle 3, -5 \rangle, \langle -1, 2 \rangle$, and $\mathbf{b} = \langle 7, 4 \rangle$. $x = 3(7) - 5(4) = 21 - 20 = 1$.
 $y = -1(7) + 2(4) = -7 + 8 = 1$. Solution: $(x, y) = (1, 1)$.

Inverse of a product

- 14 State the formula for $(AB)^{-1}$ in terms of A^{-1} and B^{-1} .
 $(AB)^{-1} = B^{-1}A^{-1}$ (note the reversed order).
- 15 If A^{-1} has rows $\langle 2, 0 \rangle, \langle 1, 1 \rangle$ and B^{-1} has rows $\langle 1, 1 \rangle, \langle 0, 1 \rangle$, find $(AB)^{-1}$.
 $(AB)^{-1} = B^{-1}A^{-1}$. Row 1: $\langle 1(2) + 1(1), 1(0) + 1(1) \rangle = \langle 3, 1 \rangle$. Row 2: $\langle 0(2) + 1(1), 0(0) + 1(1) \rangle = \langle 1, 1 \rangle$. So $(AB)^{-1}$ has rows $\langle 3, 1 \rangle, \langle 1, 1 \rangle$.
- 16 Explain briefly, using $(AB)(B^{-1}A^{-1}) = I$, why the inverse of a product must reverse the order.
 $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$ — the B 's and B^{-1} 's must sit adjacent to cancel, which only happens with the reversed order $B^{-1}A^{-1}$; the un-reversed guess $A^{-1}B^{-1}$ would not have matching factors cancel in the middle.

The transpose

- 17 For A with rows $\langle 1, 2, 3 \rangle, \langle 4, 5, 6 \rangle$ (a 2×3 matrix), find A^T .
 A^T is 3×2 with rows $\langle 1, 4 \rangle, \langle 2, 5 \rangle, \langle 3, 6 \rangle$ (the rows of A become the columns of A^T).
- 18 State the formula for $(AB)^T$ in terms of A^T and B^T .
 $(AB)^T = B^T A^T$ (transposing a product also reverses the order, just like inverses).

Symmetric matrices

- 19 A matrix A is symmetric if $A^T = A$. Is the matrix with rows $\langle 2, 3 \rangle, \langle 3, 5 \rangle$ symmetric?
 Yes: its transpose has rows $\langle 2, 3 \rangle, \langle 3, 5 \rangle$, identical to A itself.
- 20 For any matrix A , explain briefly why AA^T is always symmetric, even when A itself is not square.
 $(AA^T)^T = (A^T)^T A^T = AA^T$ (using $(XY)^T = Y^T X^T$ and $(A^T)^T = A$), so AA^T equals its own transpose — the defining property of a symmetric matrix.

Diagonal and triangular matrices

- 21 A diagonal matrix has rows $\langle 3, 0, 0 \rangle, \langle 0, -2, 0 \rangle, \langle 0, 0, 5 \rangle$. Find its inverse (recall: the inverse of a diagonal matrix is diagonal, with reciprocal entries).
 D^{-1} has rows $\langle \frac{1}{3}, 0, 0 \rangle, \langle 0, -\frac{1}{2}, 0 \rangle, \langle 0, 0, \frac{1}{5} \rangle$.
- 22 Multiply two diagonal matrices D_1 (rows $\langle 2, 0 \rangle, \langle 0, 3 \rangle$) and D_2 (rows $\langle 5, 0 \rangle, \langle 0, -1 \rangle$). What pattern do you notice?
 $D_1 D_2$ has rows $\langle 10, 0 \rangle, \langle 0, -3 \rangle$ — the product of two diagonal matrices is diagonal, with entries equal to the products of the corresponding diagonal entries.
- 23 Explain why an upper triangular matrix (all entries below the main diagonal are zero) with a 0 on its main diagonal cannot be invertible.
 The determinant of a triangular matrix equals the product of its diagonal entries. If one diagonal entry is 0, the whole product is 0, so the determinant is 0 — by the invertibility criterion, the matrix is singular.
- 24 Is the sum of two upper triangular matrices always upper triangular? Is the sum of two symmetric matrices always symmetric? Justify each briefly.
 Yes to both. Upper triangular: adding two matrices with zero entries below the diagonal keeps those entries zero. Symmetric: if $A^T = A$ and $B^T = B$, then $(A + B)^T = A^T + B^T = A + B$, so the sum is also symmetric.
- 25 Find the inverse of the upper triangular matrix with rows $\langle 2, 3 \rangle, \langle 0, 4 \rangle$ using the 2×2 inverse formula, and confirm it is also upper triangular.
 $\det = 2(4) - 3(0) = 8$. $A^{-1} = \frac{1}{8}$ times the matrix with rows $\langle 4, -3 \rangle, \langle 0, 2 \rangle$, i.e., rows $\langle \frac{1}{2}, -\frac{3}{8} \rangle, \langle 0, \frac{1}{4} \rangle$ — the lower-left entry is still 0, so A^{-1} is also upper triangular.